

# Integrated crop residue management reduces nitrogen losses by stage-specific restructuring of the soil micro-food web in dryland maize production

Author links open overlay panelJasim Iqbal <sup>a b</sup>, Zhiyuan Yao <sup>a</sup>, Carolyn-Monika Görres <sup>c</sup>, [Bo Zhu](#) <sup>a</sup>

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## Highlights

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Crop residues with synthetic fertilizers reduced total nitrogen (N) losses by 21 %.

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Stage-specific micro-food web restructuring regulated this reduction in N losses.

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Topologically important bacterial taxa predicted N losses at the seedling stage.

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Fungal diversity emerged as a significant predictor at silking and harvesting stages.

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Microbiome-informed fertilization offers a pathway to reduce environmental impacts.

## Abstract

Integrated organic-inorganic fertilization offers the potential to mitigate nitrogen (N) losses while enhancing soil biodiversity. However, a critical knowledge gap remains in understanding how soil micro-food webs regulate N dynamics across different crop growth stages and their functional linkages to seasonal N losses. To address this, we assessed four treatments from a 21-year field experiment: unfertilized control (CK), synthetic fertilizers (NPK), and partial substitution of synthetic N with pig slurry (OMNPK) or crop residues (CRNPK). We quantified total N losses via ammonia volatilization, nitrous oxide emission, and nitrate runoff and leaching, evaluated crop performance, and examined the influence of structural changes in soil micro-food webs at the seedling, silking, and

harvesting stages of maize on these processes. The results showed that OMNPK exhibited comparable total N losses to NPK (50.9 vs. 48.3 kg N ha<sup>-1</sup>). Conversely, CRNPK reduced total N losses by 21 % compared to NPK (38.0 vs. 48.3 kg N ha<sup>-1</sup>) without decreasing grain yield ( $P < 0.05$ ). Soil micro-food web analysis revealed that CRNPK enhanced the richness of phylum-level topologically important (TI) bacterial taxa (dominated by *Bacteroidota*) at the seedling stage, with a convergence in richness across treatments during the silking and harvesting stages. Additionally, stage-specific Random Forest models predicting total N losses ( $R^2 = 0.68\text{--}0.79$ ) identified TI bacterial taxa as the dominant predictor at the seedling stage, while fungal diversity emerged as a significant predictor during the silking and harvesting stages. These findings demonstrate that integrating crop residues with synthetic fertilizers effectively balances crop productivity and environmental sustainability by reducing total N losses through stage-specific restructuring of the soil micro-food web, providing an improved mechanistic understanding for managing sustainable agroecosystems.

## Introduction

Nitrogen (N) fertilization plays a crucial role in enhancing global agricultural productivity. Synthetic N inputs have been shown to increase crop yields by 55–57 %, thereby supporting food security for an expanding global population (Rosa and Gabrielli, 2023, Zhu et al., 2022). However, the widespread use of these fertilizers has led to excessive reactive N accumulation in the soil, resulting in environmental degradation and ecosystem disruption (Sabina et al., 2025). This has created a significant global N imbalance in croplands and has contributed heavily to environmental pollution (Lassaletta et al., 2014, Ludemann et al., 2024). Among the major environmental consequences, nitrate (NO<sub>3</sub><sup>-</sup>) leaching accounts for up to 73 % of N losses from croplands, threatening water quality and increasing health risks such as methemoglobinemia and cancer (Borchard et al., 2019, Wang et al., 2025). Similarly, global ammonia (NH<sub>3</sub>) volatilization from agriculture has risen by 78 % since 1980, contributing to atmospheric N deposition that causes soil acidification and aquatic ecosystem eutrophication (Gan et al., 2025). Additionally, surplus soil N can increase greenhouse gas emissions, particularly nitrous oxide (N<sub>2</sub>O), contributing to climate change (Kamran et al., 2023, Zhou et al., 2017). Managing N inputs strategically to optimize crop productivity while reducing environmental losses remains a key challenge for sustainable agriculture.

Understanding soil biological processes is crucial for addressing these N management challenges. Agricultural soils harbor complex biological communities comprising bacteria, fungi, protists, and nematodes that collectively form an integrated micro-food web governing ecosystem functionality through nutrient cycling, plant growth promotion, and

system resilience (Kiprotich et al., 2025). Within this web, bacteria and fungi serve as the primary drivers of N transformations, including mineralization of organic N compounds, nitrification of ammonium ( $\text{NH}_4^+$ ) to  $\text{NO}_3^-$ , and denitrification processes that can result in gaseous N losses (Liang et al., 2020, Sauvadet et al., 2016). Simultaneously, microbial biomass N functions as a dynamic reservoir that temporarily immobilizes N before releasing it in plant-available forms through microbial turnover (Ashraf et al., 2020, Ren et al., 2019). Soil microfauna exert critical regulatory control over these processes through direct trophic interactions, such as protist predation on bacterial populations that stimulates N mineralization, and through non-trophic mechanisms including nematode-derived metabolites that influence fungal community composition and activity (Lucas et al., 2020, Richter et al., 2019). Therefore, assessing how different fertilization alter the structure and function of these soil biological communities is essential for predicting N cycling dynamics.

Integrated organic-inorganic fertilization offer a promising strategy to reduce environmental N losses while maintaining agricultural productivity through enhanced soil biological diversity and function. Previous research demonstrates that partial substitution of synthetic N fertilizer with organic inputs such as crop residues and manure can significantly mitigate hydrological N losses compared to synthetic-only applications while preserving crop yields (Bah et al., 2020, Martinez-Feria et al., 2024). The mechanisms underlying these improvements involve organic inputs serving as carbon (C) and energy sources for diverse microbial communities, thereby stimulating soil biological activity and enhancing N cycling efficiency through increased microbial community diversity and enzymatic activity (Geisseler and Scow, 2014, Hartmann et al., 2015). Furthermore, organic matter additions increase microbial biomass C and N pools while promoting soil aggregation, which physically protects nutrients and organic matter from decomposition and reduces leaching losses (Lehmann et al., 2017, Wang et al., 2023). Despite increasing studies evaluating the impacts of integrated fertilization regimes on soil micro-food webs (Fan et al., 2025, Mosdossy et al., 2024), a critical knowledge gap remains regarding how these biological communities regulate N dynamics at different crop growth stages and their functional linkages to seasonal N losses via different pathways.

The Yangtze River Basin represents an ideal system to investigate these mechanisms. As the third largest watershed globally and home to over 400 million people, it serves as an agricultural powerhouse (Bah et al., 2020). Moreover, it is a biodiversity hotspot with fragile purple soils, facing serious challenges from soil erosion and non-point source pollution due to sloped terrain and high synthetic fertilizer use (Yao et al., 2024). These pressures threaten agroecosystem stability and downstream water quality. Therefore, identifying effective fertilization regimes that support microbial-mediated N retention while

maintaining crop productivity is essential for sustainable intensification in this region. To address these challenges, we conducted a comprehensive investigation of different integrated fertilization regimes established over a 21-year field experiment, assessing their impacts on total N losses via multiple pathways during a rain-fed maize growing season, while simultaneously characterizing microbial community responses. Our objectives were to: (1) quantify the effects of different integrated fertilization regimes on grain yield, plant N uptake, and total N losses via  $\text{NH}_3$  volatilization,  $\text{N}_2\text{O}$  emissions, and  $\text{NO}_3^-$  runoff and leaching; (2) characterize stage-specific restructuring of the soil micro-food web across seedling, silking and harvesting stages; and (3) identify functional linkages between soil micro-food web components, soil N pools, and total N losses. By integrating agronomic and ecological perspectives, this study provides essential insights into the stage-specific mechanisms that can reduce environmental trade-offs while sustaining high productivity in intensively managed subtropical agroecosystems.

### Study site

This study was conducted at the Yanting Agro-ecological Research Station of Purple Soil (31°16'N, 105°28'E) in Mianyang city, Sichuan Province, China. It is a long-term experimental site operational since 2003 as part of the Chinese Ecosystem Research Network (CERN). The site experiences subtropical monsoon conditions with annual precipitation of 826 mm and mean annual temperature of 17.3°C. The average monthly weather data for the current study period is shown in Fig. 1. The soil, classified

### N loss dynamics under different fertilization regimes

Gaseous N losses via  $\text{NH}_3$  volatilization exhibited temporal variation across treatments during early maize development (Fig. 3a). The highest  $\text{NH}_3$  flux rates were observed under OMNPK, ranging from 0.54 to 1.91 kg N ha<sup>-1</sup> d<sup>-1</sup>, followed by CRNPK with fluxes between 0.52 and 1.48 kg N ha<sup>-1</sup> d<sup>-1</sup>.  $\text{N}_2\text{O}$  emissions demonstrated a similar temporal pattern, with peak fluxes occurring within the first two weeks post-fertilization (Fig. 3b). OMNPK generated markedly higher  $\text{N}_2\text{O}$  fluxes throughout the growing

### Differential N loss pathways under fertilization regimes

$\text{NH}_3$  volatilization was the dominant pathway of gaseous N losses, with OMNPK showing the highest fluxes among treatments. This increase is attributable to rapid urea hydrolysis in pig slurry, where urinary urea is enzymatically broken down by urease in fecal matter (Hao et al., 2019). CRNPK also elevated  $\text{NH}_3$  fluxes compared with NPK, likely due to increased availability of urea and organic N from crop residues, which stimulated ureolytic activity (Moharana and Biswas, 2022). The enhanced urease

### Conclusions

This study demonstrates that partial substitution of synthetic N with crop residues (CRNPK) reduced total N losses across entire maize growth season by 21 % compared to NPK, while maintaining comparable grain yield. In contrast, partial synthetic N substitution with pig slurry (OMNPK) produced total N losses similar to NPK, driven mainly by higher gaseous emissions. Our data indicate that the total N losses reduction under CRNPK stems from stage-specific correlations within the soil micro-food

#### CRedit authorship contribution statement

**Bo Zhu:** Writing – review & editing, Validation, Supervision, Project administration, Funding acquisition, Conceptualization. **Carolyn-Monika Görres:** Writing – review & editing, Validation, Methodology, Data curation. **Zhiyuan Yao:** Writing – review & editing, Validation, Supervision, Methodology, Funding acquisition, Data curation, Conceptualization. **Jasim Iqbal:** Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization.

#### Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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